

Systems Science & Control Engineering

An Open Access Journal

ISSN: 2164-2583 (Online) Journal homepage: www.tandfonline.com/journals/tssc20

Digital twin-enabled real-time parameter estimation of a wheeled mobile robot

Shinta R. Kosmaga , Augie Widyotriatmo , Estiko Rijanto & Agus Hasan

To cite this article: Shinta R. Kosmaga , Augie Widyotriatmo , Estiko Rijanto & Agus Hasan (2026) Digital twin-enabled real-time parameter estimation of a wheeled mobile robot, Systems Science & Control Engineering, 14:1, 2634459, DOI: 10.1080/21642583.2026.2634459

To link to this article: https://doi.org/10.1080/21642583.2026.2634459

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 25 Feb 2026.

Submit your article to this journal

Article views: 578

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tssc20

SYSTEMS SCIENCE & CONTROL ENGINEERING: AN OPEN ACCESS JOURNAL
2026, VOL. 14, NO. 1, 2634459
https://doi.org/10.1080/21642583.2026.2634459

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Digital twin-enabled real-time parameter estimation of a wheeled mobile robot

Shinta R. Kosmaga[a], Augie Widyotriatmo[b], Estiko Rijanto[a] and Agus Hasan[c]

[a]Research Center for Smart Mechatronics, Badan Riset dan Inovasi Nasional (BRIN), Bandung, Indonesia; [b]Instrumentation and Control Research Group, Institut Teknologi Bandung, Bandung, Indonesia; [c]Department of ICT and Natural Sciences, Norwegian University of Science and Technology, Ålesund, Norway

ABSTRACT

This paper presents a real-time parameter estimation framework for wheeled mobile robots (WMR) using a digital twin architecture based on the Wide-Array of Nonlinear Dynamics Approximation (WyNDA). A WyNDA-based adaptive observer is applied to perform static and time-varying parameter estimation through simulations and experiments, with comparative evaluation against SINDy. Simulation results show that WyNDA consistently estimates both static and time-varying parameters with a MAPE below 6.1%, while experimental validation yields a MAPE of approximately 10%, outperforming SINDy in both cases. Furthermore, the experimental results are implemented in a digital twin, which achieves accurate position tracking during data loss, with errors of 3% and 6% in the x and y estimates, respectively. These results demonstrate WyNDA's effectiveness for real-time parameter estimation and its suitability for digital twin applications under incomplete sensing conditions.

ARTICLE HISTORY
Received 22 August 2025
Accepted 12 February 2026

KEYWORDS
Parameter estimation; nonlinear systems; wheeled mobile robot; digital twin

1. Introduction

Wheeled mobile robots (WMRs) are widely used in logistics, surveillance, and manufacturing due to their mechanical simplicity and high mobility. As these systems increasingly operate in dynamic and uncertain environments, the demand for autonomy has spurred significant developments in decision-making algorithms and control strategies. Achieving robust autonomous behaviour under such conditions requires accurate and timely knowledge of both the internal and external states of the robot. Although onboard sensors provide partial state information, there are many scenarios where direct measurements are unavailable or unreliable due to sensor limitations, environmental interference, or hardware failures.

A promising approach to overcoming these limitations is the use of digital twins, which offer a real-time virtual representation of physical systems that continuously synchronizes with its real-world counterpart (Veledar et al., 2019). A digital twin is defined as a dynamic, data-driven model that maintains a bidirectional interaction with its physical system, enabling simulation, monitoring, and prediction based on real-time data (Fuller et al., 2020). In mobile robotics, digital twins are increasingly employed to support performance optimization, predictive maintenance, and fault diagnosis by closely replicating the system's behaviour in simulation (Piromalis & Kantaros, 2022). However, the accuracy of a digital twin depends on the fidelity of the underlying dynamic model and the correctness of its parameter values.

Operational conditions such as payload changes, surface irregularities, and wheel slip continuously affect these parameters (Hong et al., 2015). As a result, static modelling approaches alone are insufficient to ensure robust performance under realistic operating conditions. Consequently, an adaptive estimation strategy capable of tracking such variations in real time is required to maintain an accurate representation of system dynamics. In this context, real-time parameter estimation has emerged as a key enabler for adaptive digital twins, enabling continuous model refinement based on incoming sensor data and enhancing the system's responsiveness to environmental and operational changes (Zarringhalam et al., 2012).

CONTACT Agus Hasan agus.hasan@ntnu.no

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

1.1. Related works

This study focuses on a car-like wheeled mobile robot. Although the platform is a scaled-down robotic system, its kinematic and dynamic behaviour closely resembles that of full-scale ground vehicles. Consequently, techniques developed for vehicle parameter estimation remain applicable and serve as the methodological foundation for this research. Numerous studies have explored vehicle parameter estimation using various approaches, with model-based methods being among the most widely adopted. These methods rely on mathematical models derived from fundamental physical principles to describe system dynamics.

For instance, Hong et al. (2015) employed a dual unscented Kalman filter (DUKF) to estimate vehicle mass, the distance from the center of gravity (CoG) to the front axle, and the yaw moment of inertia. Xu et al. (2015) applied a dual extended Kalman filter (DEKF) to estimate road friction coefficients and vehicle state variables. Similarly, Jin et al. (2019) utilized DEKF for estimating mass, yaw moment of inertia, and state variables. Li et al. (2021) implemented DUKF to estimate several vehicle parameters, including the distance from the CoG to the front axle, CoG height, slip ratio gradient, yaw and roll moments of inertia, and tyre cornering stiffness. In addition, Oei and Sawodny (2022) estimated vehicle mass and frictional forces using a disturbance observer-based Kalman filter. More recently, Jin et al. (2023) proposed a method based on the dual central difference Kalman filter (DCDKF) to estimate vehicle mass, the distance from the CoG to the front axle, the yaw moment of inertia, and vehicle state variables.

Despite their widespread use, model-based methods often face significant challenges, particularly when applied to complex systems or scenarios where accurate modelling is difficult to achieve. As an alternative, data-driven estimation methods have gained increasing attention as a means to overcome these limitations. These approaches utilize historical data or simulation results to estimate system parameters and state variables without relying on an explicit physical model.

For example, Dong et al. (2021) applied a recurrent neural network (RNN) to estimate lateral and longitudinal adhesion coefficients, along with vehicle state variables. Similarly, Kim and Jung (2024) employed an artificial neural network (ANN) to estimate vehicle mass using perturbed input data, demonstrating the capability of neural networks to handle non-ideal or noisy data.

Nevertheless, data-driven approaches such as machine learning generally treat the system as a black box, which limits the ability to analyze and understand internal dynamics. This poses specific challenges in terms of model interpretability and validation. To address these limitations while leveraging the strengths of both paradigms, recent studies have proposed hybrid methods that combine physics-based models with machine learning algorithms. For instance, Gräber et al. (2019) integrated an RNN with a vehicle kinematic model to improve the accuracy of lateral sideslip angle estimation. Likewise, Yu et al. (2022) proposed a method combining a feedforward neural network (FFNN) with recursive least squares (RLS) to estimate vehicle mass, aiming to achieve adaptive estimation while preserving the physical interpretability of the system.

Among interpretable data-driven approaches, one notable method is the Sparse Identification of Nonlinear Dynamics (SINDy) (Brunton et al., 2016). SINDy uses sparse regression to identify parsimonious governing equations from time-series data by selecting the most relevant nonlinear terms from a predefined library (Champion et al., 2019). It is particularly effective for discovering interpretable dynamics from high-dimensional datasets, making it a powerful tool in data-driven modelling. However, a key limitation of SINDy lies in its sensitivity to measurement noise, as it typically requires clean or low-noise data, an assumption that is often difficult to meet in real-world applications.

Building on these diverse strands of research, Hasan (2024a, 2024b) introduced a novel framework for mathematical model identification of dynamic systems, known as Wide-Array of Nonlinear Dynamics Approximation (WyNDA). An adaptive observer algorithm is employed to address the estimation challenges inherent in the identification process. By utilizing empirical data, the WyNDA framework is applied to derive equations that accurately represent system dynamics, closely approximating models traditionally obtained through physics-based modelling approaches (Hasan & Hilmi, 2024). Extending this framework, WyNDA has also been applied to estimate piecewise time-varying vehicle parameters induced by varying seat arrangements under a constant mass condition (Kosmaga et al., 2025).

1.2. Contribution

This study employs the WyNDA framework for real-time estimation of key WMR parameters, including mass, yaw moment of inertia, the longitudinal position of the center of gravity (CoG), and the cornering stiffness of both front and rear tyres. Although vehicle parameter estimation methods have been extensively investigated in the literature, their applicability to highly nonlinear WMR dynamics remains limited, as summarized as follows: (1) Physical model-based methods with Kalman filters require specific excitation conditions – such as predefined speed ranges or steering angle thresholds – to ensure convergence, conditions that may not be consistently satisfied in real-world driving scenarios (Hong et al., 2015; Jin et al., 2019; Li et al., 2021; Oei & Sawodny, 2022). They also incorporate linearization techniques to simplify the system dynamics, potentially leading to the loss of important nonlinear characteristics (Jin et al. 2019, 2023; Xu et al., 2015). (2) Data-driven or learning-based methods rely on offline training which inherently restrict their capability for real-time parameter estimation (Brunton et al., 2016; Dong et al., 2021; Gräber et al., 2019; Kim & Jung, 2024; Yu et al., 2022). (3) None of the aforementioned studies estimated time-varying parameter values with validation from simulation or experimental data.

This study addresses the aforementioned limitations of time-varying parameter estimation by implementing the WyNDA framework and proposing a digital twin for a WMR. The specific contributions of this study are as follows:

(1) Capturing nonlinear vehicle dynamics by estimating WMR parameters without linearization.
(2) Enabling real-time parameter estimation, allowing parameter variations during WMR operation.
(3) Conducting simulations and experimental validations on a WMR to estimate its key parameter values. By incorporating time-varying parameter, these simulations complement existing literature on the WyNDA framework, which has previously relied on constant parameters for non-ground vehicles (Hasan, 2024a, 2024b; Hasan & Hilmi, 2024) and for ground vehicles (Kosmaga et al., 2024). Moreover, this work extends previous study through a comprehensive comparison with SINDy and validation using physical experiments (Kosmaga et al., 2025).
(4) Presenting a digital twin that integrates physical WMR operation with real-time simulation. It supports reliable monitoring and control, thereby potentially enhancing the performance and dependability of autonomous WMR systems.

The overall system architecture is illustrated in Figure 1, highlighting the real-time interaction between the physical system, the estimation module, and the digital twin.

1.3. Organization

The remainder of this paper is structured as follows. Section 2 introduces the WyNDA methodology, outlining its core principles and estimation framework. Section 3 details the application of WyNDA to a WMR system, including the formulation of the WMR dynamics model and its integration within the estimation process. Section 4 presents the results of the study, encompassing numerical simulations with static and piecewise time-varying parameter estimation, experimental validation, and digital twin implementation. Finally, Section 5 concludes the paper and discusses potential directions for future research.

2. Overview of WyNDA framework

In contrast to conventional machine learning approaches, which often model systems as black boxes, WyNDA is capable not only of predicting future system states but also of constructing interpretable mathematical representations of the system's dynamics. This interpretability allows for more transparent analysis and a better understanding of the underlying physical behaviour, making WyNDA particularly suitable for applications that require both accuracy and insight into system structure.

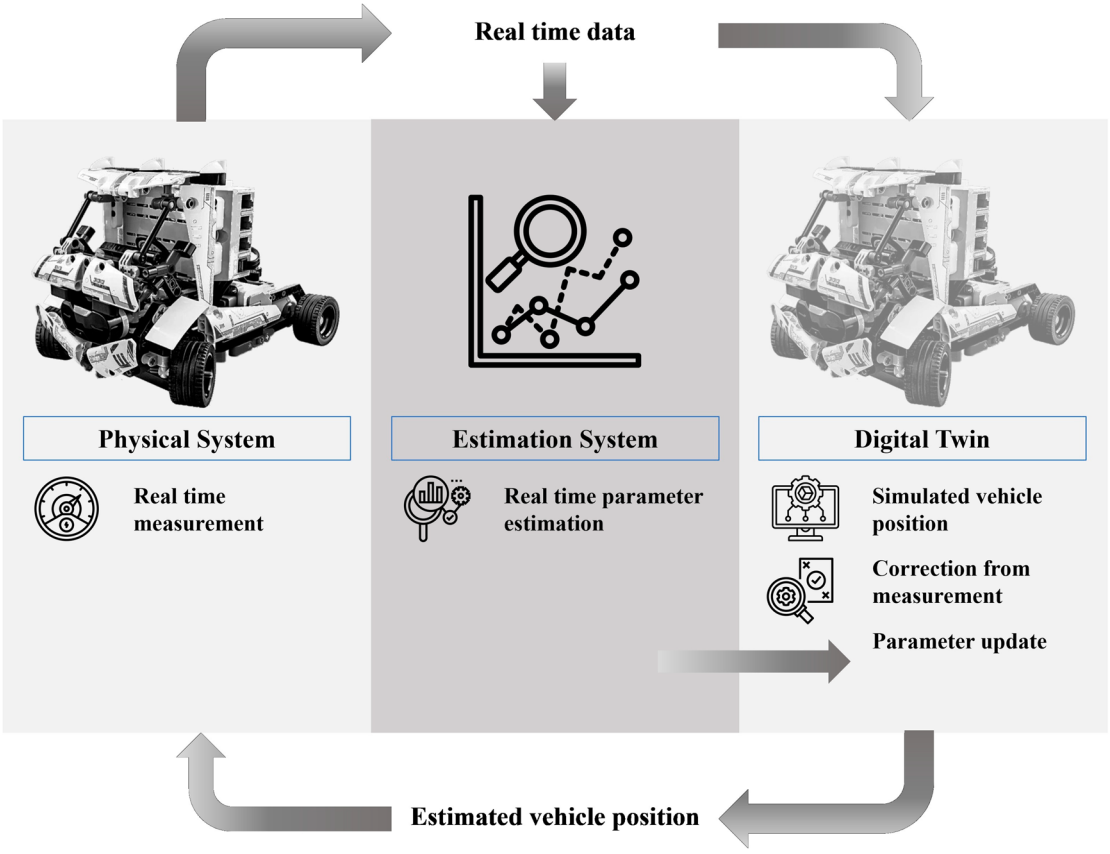


Figure 1. System architecture for digital twin development.

The continuous dynamical system is describe by the following equation.

$$\dot{\boldsymbol{x}}(t) = f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{\theta}_p) \tag{1}$$

$$\boldsymbol{x}(t_0) = \boldsymbol{x_0}. \tag{2}$$

The evolution of the current state over time, denoted by $\dot{\boldsymbol{x}}$, can be expressed as a function of the current state, control input, and system parameters. In this study, the function f is assumed to be known through system modelling, the state vector $\boldsymbol{x}$ is obtained from measurements, and the control input $\boldsymbol{u}$ is considered known. However, the parameter vector $\boldsymbol{\theta_p}$ is typically unknown or difficult to measure directly. The measurement from sensors are recorded as discrete data $\boldsymbol{y}(k)$ where the $\eta(k)$ represent the measurement noise or uncertainty, such that

$$\boldsymbol{y}(k) = \boldsymbol{x}(k) + \eta(k). \tag{3}$$

To estimate the unknown parameters, WyNDA involves three key steps: constructing an approximation model, applying an adaptive observer algorithm, and converting the discrete-time approximation into a continuous-time model (Kosmaga et al., 2024).

(1) Approximation model
WyNDA approximates the system evolution in discrete time as

$$\boldsymbol{x}(k+1) = \boldsymbol{x}(k) + \boldsymbol{\Lambda}(\boldsymbol{y}(k), \boldsymbol{u}(k))\boldsymbol{\theta}, \tag{4}$$

where $\boldsymbol{\Lambda}(\boldsymbol{y}(k), \boldsymbol{u}(k))$ is an approximation function composed of a wide array of basis functions, and $\boldsymbol{\theta}$ is a parameter vector to be estimated. This formulation can be compare with Euler discretization so the approximation model can be intrepreted as

$$\boldsymbol{\Lambda}(\boldsymbol{y}(k), \boldsymbol{u}(k))\boldsymbol{\theta} \approx \Delta t f(\boldsymbol{x}(k), \boldsymbol{u}(k), \boldsymbol{\theta}_p), \tag{5}$$

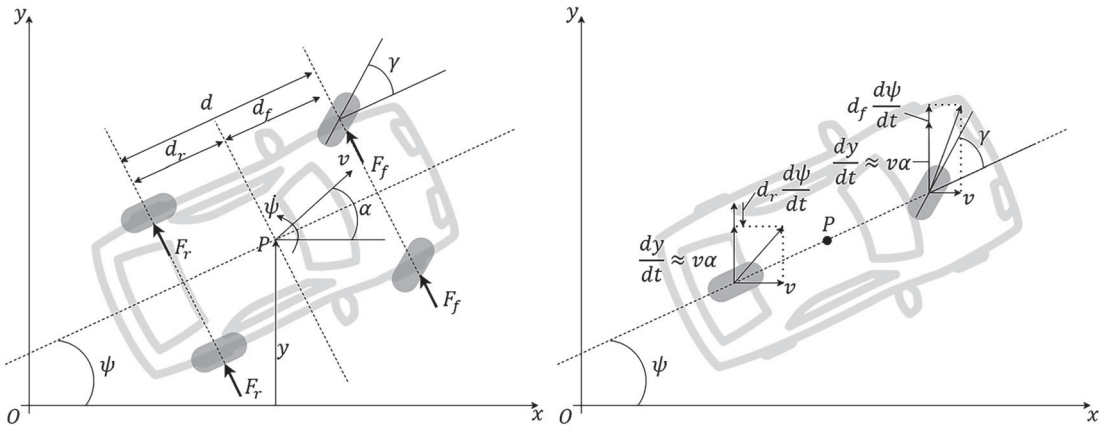


Figure 2. Schematic of WMR Dynamics Modelling: WMR in a coordinate system fixed to the ground, and the simplified model.

(2) Adaptive observer algorithm
The unknown parameter vector $\boldsymbol{\theta}$ is estimated using an adaptive observer that jointly updates the state and parameter estimates based on measurement feedback. The observer gains are computed from estimation error covariances and tuning matrices, ensuring stable convergence under measurement noise and modelling uncertainties. Forgetting factors are incorporated to accommodate slowly varying parameters.

(3) Once the parameters $\boldsymbol{\theta}$ are estimated, the continuous-time dynamics can be reconstructed using (5), enabling the identification of system parameters.

Further details on the WyNDA framework are available in Hasan (2024b).

3. Implementation on WMR system

3.1. WMR dynamics model

In this study, a two degrees-of-freedom (2-DOF) WMR dynamics model is adopted in a global coordinate system, as described in Abe (2015) and Rajamani (2012). The analysis focuses solely on lateral and yaw motions, which represent the WMR's behaviour in the horizontal plane.

Figure 2 illustrates the simplified WMR model known as the single-track or bicycle model, where the front and rear wheel pairs are approximated by a single front and a single rear wheel, respectively (Qin et al., 2022). The point labelled P represents the WMR's center of gravity, where d_f and d_r denote the distances from the CoG to the front and rear axles. The angle between the WMR's longitudinal direction and the x-axis, known as the yaw angle, is denoted by ψ, while the angle between the WMR's heading direction and the x-axis is denoted by α. The WMR velocity is represented by v, the steering angle by γ, and the lateral displacement by y. The lateral forces acting on the front and rear wheels are denoted by F_f and F_r, respectively.

Under the small-angle assumption, the lateral and yaw dynamics are governed by:

$$m\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = 2F_f + 2F_r. \tag{6}$$

$$I_y\frac{\mathrm{d}^2 \psi}{\mathrm{d}t^2} = 2d_f F_f - 2d_r F_r \tag{7}$$

where m is the WMR mass and I_y is the yaw moment of inertia. The lateral forces on the front and rear wheels, F_f anf F_r, can be expressed as:

$$F_f = -C_f\beta_f = -C_f\left(\frac{1}{v}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{d_f}{v}\frac{\mathrm{d}\psi}{\mathrm{d}t} - \psi - \gamma\right), \tag{8}$$

$$F_r = -C_r\beta_r = -C_r\left(\frac{1}{v}\frac{\mathrm{d}y}{\mathrm{d}t} - \frac{\mathrm{d}_r}{v}\frac{\mathrm{d}\psi}{\mathrm{d}t} - \psi\right). \tag{9}$$

where C_f and C_r denote the cornering stiffness of the front and rear tyres.

From Equations (6)–(9), the complete WMR equations of motion are summarized as follows:

$$\dot{x} = v\cos(\beta + \psi) \tag{10a}$$

$$\dot{y} = v\sin(\beta + \psi) \tag{10b}$$

$$\dot{\psi} = r \tag{10c}$$

$$\ddot{y} = \frac{2(C_f + C_r)}{m}\psi - \frac{2(C_f + C_r)}{mv}\dot{y} - \frac{2(d_f C_f - d_r C_r)}{mv}\dot{\psi} + \frac{2C_f}{m}\gamma \tag{10d}$$

$$\ddot{\psi} = \frac{2(d_f C_f - d_r C_r)}{I_y}\psi - \frac{2(d_f C_f - d_r C_r)}{I_y v}\dot{y} - \frac{2(d_f^2 C_f + d_r^2 C_r)}{I_y v}\dot{\psi} + \frac{2d_f C_f}{I_y}\gamma \tag{10e}$$

where $\boldsymbol{x} = [x \;\; y \;\; \psi \;\; \dot{y} \;\; \dot{\psi}]^{\mathsf{T}}$ represents the WMR state variable, and $\boldsymbol{u} = [v \;\; \gamma]^{\mathsf{T}}$ represents the input.

The parameters m, I_y, d_f, d_r, C_f, dan C_r represent internal characteristics of the WMR that govern its dynamic behaviour. These parameters are typically not directly measurable and may vary over time due to payload changes or surface irregularities. In this study, the cornering stiffness coefficients C_f and C_r are treated as known values, while the remaining parameters, namely the WMR mass m, yaw moment of inertia I_y, and the longitudinal distances from the center of gravity to the front and rear axles (d_f and d_r), will be estimated using the WyNDA framework.

3.2. Approximation function design

The construction of the approximation function is a critical component in the implementation of WyNDA. This function consists of a wide array of basis functions Ω, which can be defined using various strategies. If the selected basis functions fail to adequately represent the system dynamics, the resulting approximation function may approach zero, thereby diminishing estimation accuracy.

In this study, five basis functions Ω_j are employed, each corresponding to a state derivative in (10a). The basis functions are given by:

$$\boldsymbol{\Omega}_1(\boldsymbol{y}(k), \boldsymbol{u}(k)) = [u_1(k)\cos(\beta + y_3(k))] \tag{11a}$$

$$\boldsymbol{\Omega}_2(\boldsymbol{y}(k), \boldsymbol{u}(k)) = [u_1(k)\sin(\beta + y_3(k))] \tag{11b}$$

$$\boldsymbol{\Omega}_3(\boldsymbol{y}(k), \boldsymbol{u}(k)) = [y_5(k)] \tag{11c}$$

$$\boldsymbol{\Omega}_4(\boldsymbol{y}(k), \boldsymbol{u}(k)) = \begin{bmatrix} y_3(k) & \dfrac{y_4(k)}{u_1(k)} & \dfrac{y_5(k)}{u_1(k)} & u_2(k) \end{bmatrix} \tag{11d}$$

$$\boldsymbol{\Omega}_5(\boldsymbol{y}(k), \boldsymbol{u}(k)) = \begin{bmatrix} y_3(k) & \dfrac{y_4(k)}{u_1(k)} & \dfrac{y_5(k)}{u_1(k)} & u_2(k)] \end{bmatrix} \tag{11e}$$

Based on (3), $\boldsymbol{y}(k)$ corresponds to the measured WMR state vector with additive measurement noise. Accordingly, the complete approximation function is expressed as:

$$\boldsymbol{\Lambda}(*) = \begin{bmatrix} \boldsymbol{\Omega}_1(*) & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Omega}_2(*) & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Omega}_3(*) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{\Omega}_4(*) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{\Omega}_5(*) \end{bmatrix} \tag{12}$$

with $(*)$ represents $(\boldsymbol{y}(k), \boldsymbol{u}(k))$. The parameter vector was defined as follows:

$$\boldsymbol{\theta} = [\theta_1 \quad \theta_2 \quad \cdots \quad \theta_{10} \quad \theta_{11}]^{\mathsf{T}}. \tag{13}$$

The estimation of internal WMR parameters becomes feasible after the adaptive observer yields the state vector $\bar{\boldsymbol{x}}$ and the parameter vector $\bar{\boldsymbol{\theta}}$. By leveraging the system dynamics approximation described in (5), the

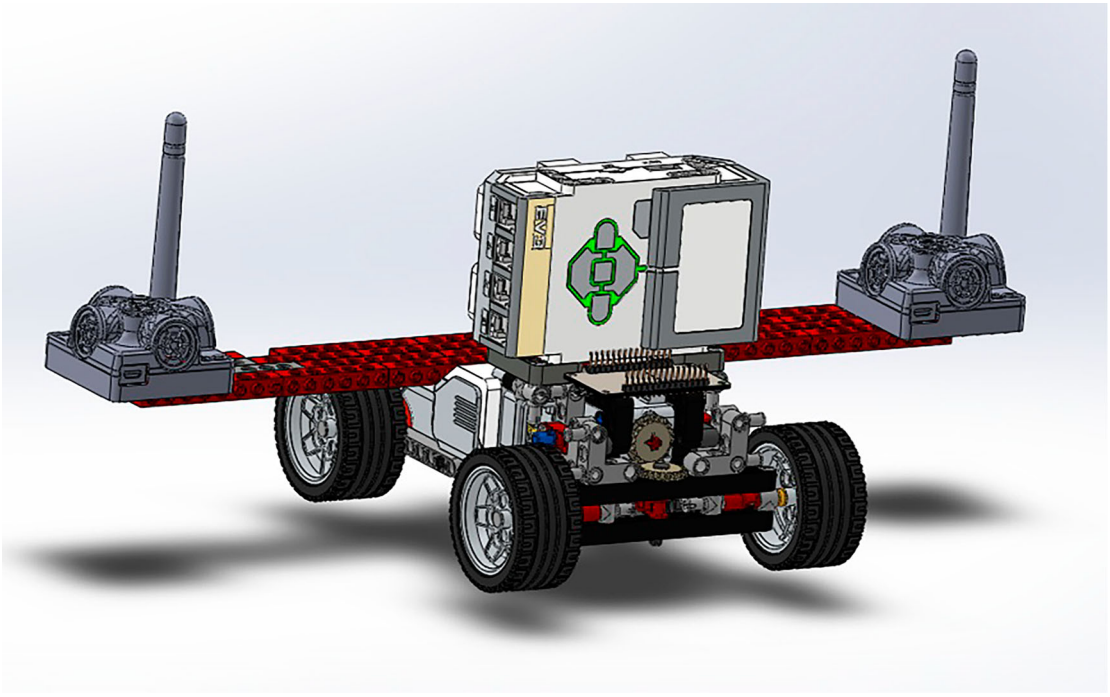


Figure 3. 3D model of the car-like WMR used in the experiment.

Table 1. Reference value of WMR parameters.

Parameter	Value	Parameter	Value	Parameter	Value
m (kg)	0.856	d_f (m)	6.87×10^{-2}	C_f (N/rad)	4×10^{-2}
I_y (kgm^2)	7.94×10^{-3}	d_r (m)	6.73×10^{-2}	C_r (N/rad)	4.3×10^{-2}

estimated physical parameter vector $\bar{\boldsymbol{\theta}}_p$ can be derived as follows:

$$\bar{m} = \frac{2C_f \Delta t}{\bar{\theta}_7} \tag{14a}$$

$$\bar{d}_f = \frac{2C_r d - \frac{\bar{\theta}_6 m}{\Delta t}}{2(C_f + C_r)} \tag{14b}$$

$$\bar{d}_r = d - d_f \tag{14c}$$

$$\bar{I}_y = \frac{2C_f d_f \Delta t}{\bar{\theta}_{11}} \tag{14d}$$

This formulation is used to compute the physical parameter estimates once the approximation parameter vector $\bar{\boldsymbol{\theta}}$ has been obtained.

4. Experimental result and discussion

This section presents both numerical simulations and physical experiments. The physical platform used in this study is a car like wheeled mobile robot. A 3D model of the WMR, shown in Figure 3, is constructed to estimate key internal parameters, such as mass, center of gravity (CoG) position, and yaw moment of inertia, which serve as reference values for comparison. Meanwhile, the front and rear cornering stiffness, assumed to be known a priori, are derived from car-like vehicle data (Abe, 2015) using the Buckingham π theorem (Verma et al., 2008). The complete set of internal parameters for the physical WMR is summarized in Table 1.

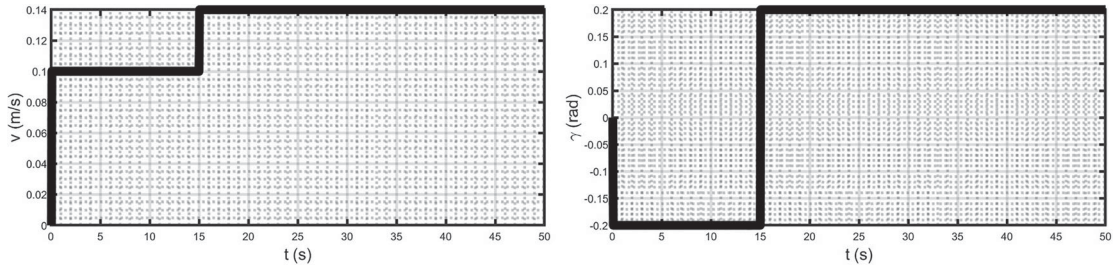


Figure 4. Simulation input signals for constant-parameter simulation.

Table 2. Initial conditions of the adaptive observer parameters for the WMR constant-parameter simulation.

Parameter	Value	Parameter	Value	Parameter	Value
$\boldsymbol{E_x}(0\mid -1)$	$10\boldsymbol{I}$	$\boldsymbol{T_x}(1)$	$1\boldsymbol{I}$	$\boldsymbol{\Pi}(0\mid 0)$	0
$\boldsymbol{E_\theta}(0\mid -1)$	$1\boldsymbol{I}$	$\boldsymbol{T_\theta}(1)$	$1\boldsymbol{I}$	$\mu_{\boldsymbol{x}}$	0.9999
				$\mu_{\boldsymbol{\theta}}$	0.9995

Table 3. Estimated WMR parameters and corresponding RMSE and MAPE for the constant-parameter simulation.

Parameter	Reference Value	Estimated Value	RMSE	MAPE
m (kg)	0.856	0.842	0.0143	1.63%
I (kgm^2)	7.94×10^{-3}	7.7×10^{-3}	2.65×10^{-4}	3.31%
d_f (m)	6.87×10^{-2}	6.91×10^{-2}	4.09×10^{-4}	0.53%
d_r (m)	6.73×10^{-2}	6.69×10^{-2}	4.09×10^{-4}	0.54%

4.1. Numerical simulation

4.1.1. Static parameter estimation

The simulation spanned 50 s with a sampling period of 0.05 s, yielding 1000 data points. For real-time parameter estimation, the WyNDA was run in parallel with the WMR dynamics model. The input profiles used in the simulation are presented in Figure 4. To account for measurement uncertainty, noise was introduced into the system outputs to reflect realistic sensing conditions.

The performance of the adaptive observer depends on several hyper-parameters, including the tuning matrices $\boldsymbol{T_x}$ and $\boldsymbol{T_\theta}$, the forgetting factors $\mu_{\boldsymbol{x}}$ and $\mu_{\boldsymbol{\theta}}$, and the initial error covariance matrices $\boldsymbol{E_x}$ and $\boldsymbol{E_\theta}$. This initial parameter settings are summarized in Table 2.

Figure 5 visualizes the WMR trajectory during the simulation. The black line represents the trajectory obtained from the WMR dynamics model, while the red dot line shows the trajectory inferred via the WyNDA-based estimation method. As seen in the figure, the two trajectories overlap almost entirely, demonstrating the high positional accuracy achieved by the WyNDA framework.

Estimated WMR parameters, including mass, yaw moment of inertia, and distances from the center of gravity to the front and rear axles, are shown in Figure 6, which illustrates their convergence over time. The results indicate that the parameter estimates converge after approximately 15 s, corresponding to about 300 data points.

The detailed results of the WMR parameter estimation are presented in Table 3 To minimize the influence of transient fluctuations, the final parameter estimates are computed using the last 150 samples. The root mean square error (RMSE) and mean absolute percentage error (MAPE) values in the Table 3 indicate how closely WyNDA is able to approximate the parameter values obtained from the 3D model. The low RMSE values across all parameters demonstrate that WyNDA delivers high-precision estimation of the WMR's internal characteristics under simulated conditions.

Furthermore, the simulated WMR dynamics data were recorded and subsequently used for parameter estimation using SINDy. Although SINDy is not a real-time estimation method, it is a widely used data-driven approach for system identification that aims to recover governing equations from measured data. SINDy can

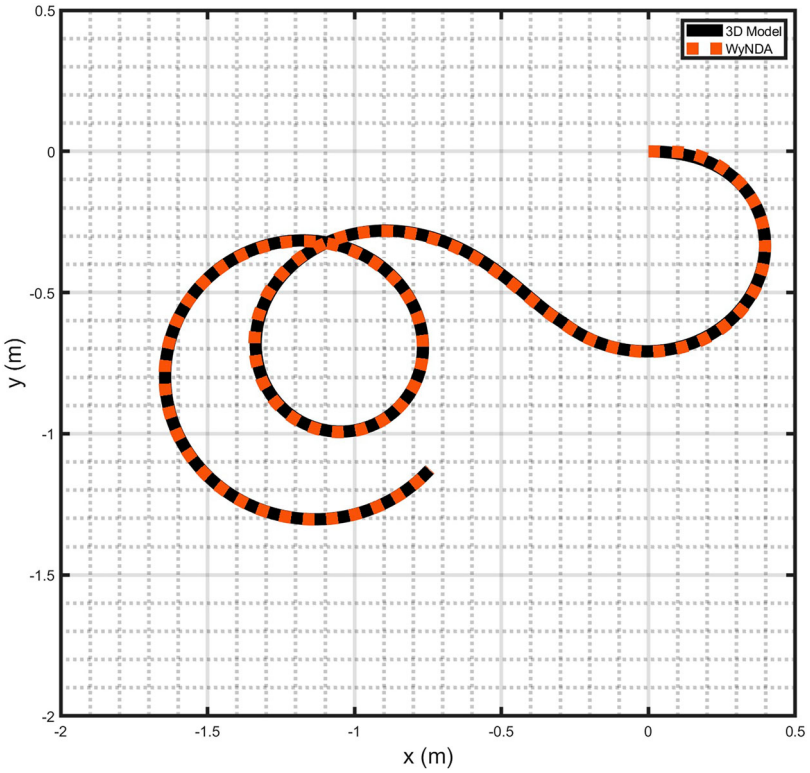


Figure 5. Simulated WMR trajectory under constant-parameter simulation.

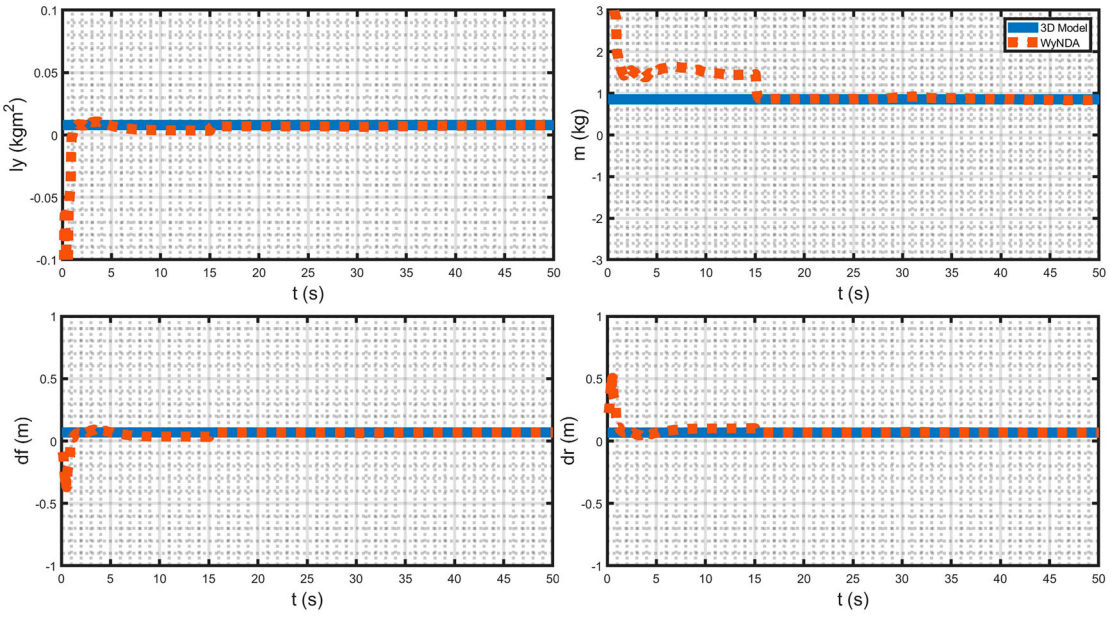


Figure 6. WMR parameter estimates during constant-parameter simulation.

be applied under the same simulation settings as WyNDA, including identical input signals, target parameters, and noise conditions. This allows SINDy to perform parameter estimation under comparable operating scenarios.

Comparative evaluation with SINDy is conducted by examining the number of data observations required by each method to achieve an estimation accuracy of 80%, with the parameter-wise comparison summarized in Table 4. In addition, time histories of the parameter estimation results for both WyNDA and SINDy over five

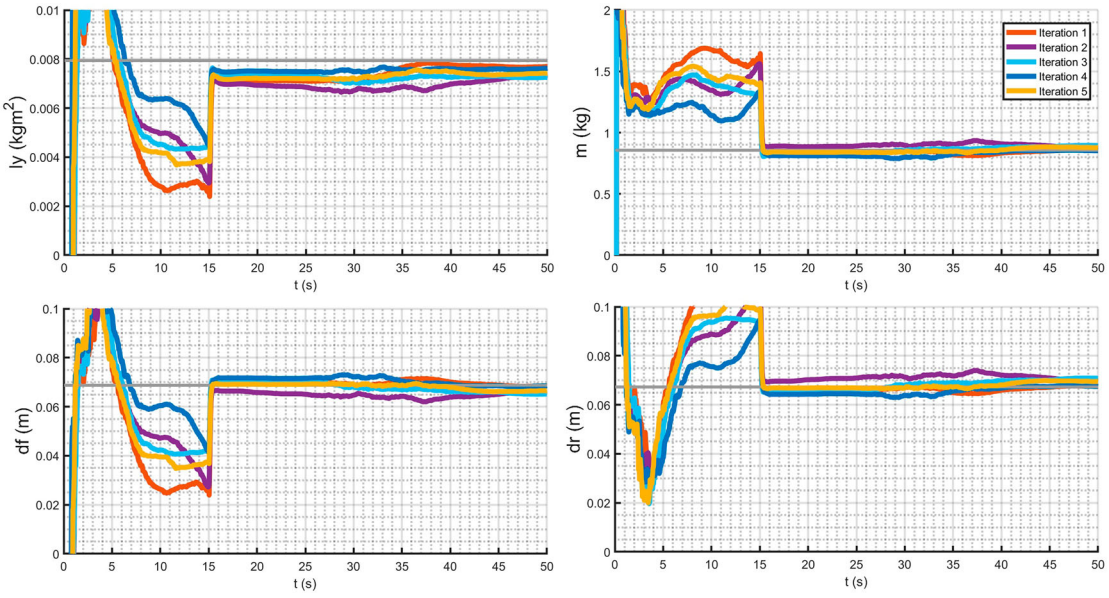


Figure 7. WyNDA-based parameter estimation results over five simulation runs.

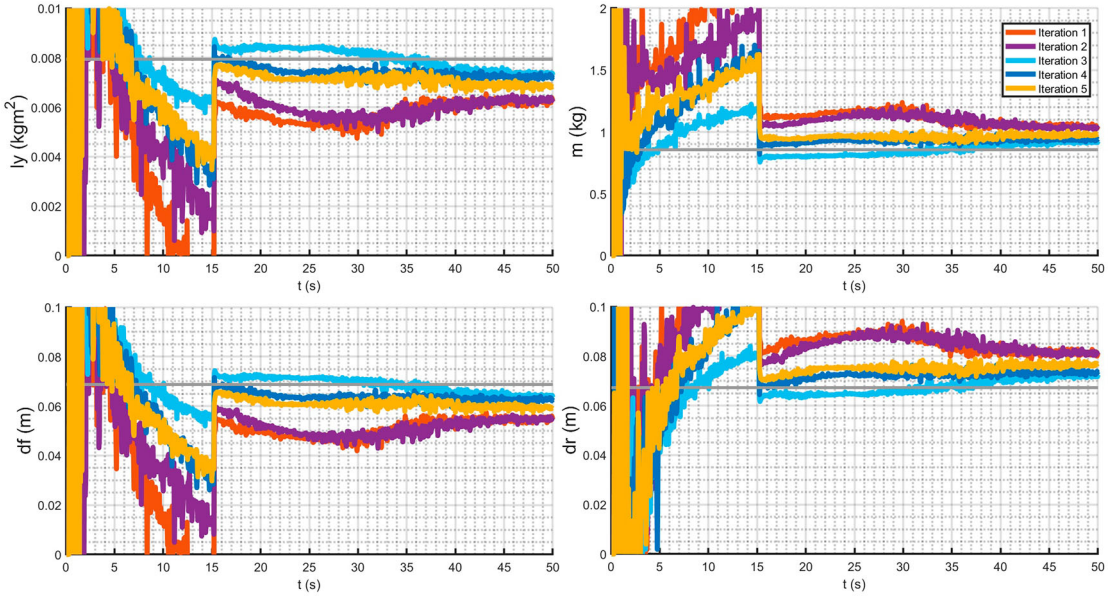


Figure 8. SINDy-based parameter estimation results over five simulation runs.

simulation runs are presented in Figures 7 and 8 to illustrate their convergence behaviour. The results indicate that WyNDA reaches the 80% accuracy threshold after approximately 304–306 data samples. Under nominal noise conditions, SINDy requires a comparable amount of historical data to achieve a similar level of estimation accuracy. However, SINDy is found to be more sensitive to measurement noise; across repeated simulation runs with different noise realizations, the method does not consistently achieve the 80% accuracy threshold, even when the full simulation horizon of 50 s (corresponding to 1000 data points) is available.This observation is further supported by quantitative error metrics: at a simulation horizon of 50 s, the RMSE $\pm$ standard deviation of the mass estimation achieved by WyNDA and SINDy is 0.02 ± 0.015 and 0.127 ± 0.058, respectively. These results demonstrate that WyNDA outperforms SINDy in estimating the WMR's mass under constant parameter conditions.

Table 4. Comparison of Data Requirements to Achieve 80% Estimation Accuracy for WyNDA and SINDy.

	WyNDA				SINDy			
Iteration	m	I	d_f	d_r	m	I	d_f	d_r
1	306	306	305	305	> 1000	> 1000	> 1000	> 1000
2	305	305	305	305	> 1000	> 1000	> 1000	> 1000
3	304	304	304	304	306	306	298	298
4	304	305	304	304	306	306	306	306
5	305	305	305	305	306	307	306	306

Table 5. WMR parameter variations under different payload-change scenarios.

Scenario	m	I	d_f	d_r
1	0.942	8.3×10^{-3}	6.93×10^{-2}	6.67×10^{-2}
2	1.112	10.08×10^{-3}	7.86×10^{-2}	5.74×10^{-2}
3	1.284	10.35×10^{-3}	8.275×10^{-2}	5.325×10^{-2}

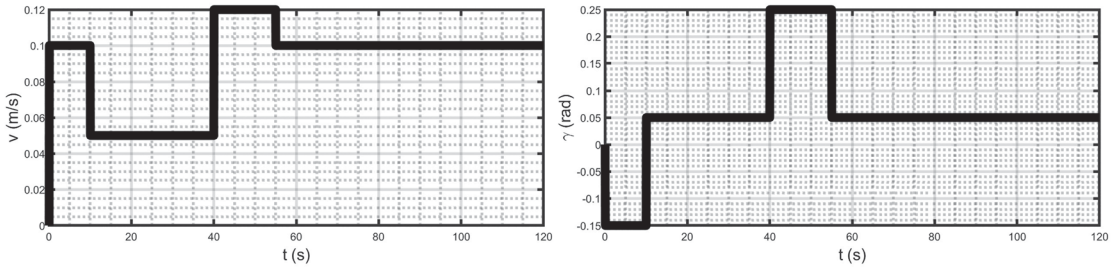


Figure 9. Input signals for the WMR parameter-change simulation.

Table 6. Initial conditions of the adaptive observer parameters for the WMR parameter-change simulation.

Parameter	Value	Parameter	Value	Parameter	Value
$\boldsymbol{E_x}(0 \mid -1)$	$1 \times 10^5 \boldsymbol{I}$	$\boldsymbol{T_x}(1)$	$8.93 \times 10^4 \boldsymbol{I}$	$\boldsymbol{\Pi}(0 \mid 0)$	0
$\boldsymbol{E_\theta}(0 \mid -1)$	$3.75 \times 10^4 \boldsymbol{I}$	$\boldsymbol{T_\theta}(1)$	$1 \times 10^5 \boldsymbol{I}$	$\mu_{\boldsymbol{x}}$	0.995
				$\mu_{\boldsymbol{\theta}}$	0.9964

4.1.2. Piecewise time-varying parameter

The simulation was conducted for 120 s with a sampling period of 0.05 s. To simulate piecewise time-varying parameters, the WMR was first operated with its nominal parameters, after which an additional load was applied at $t = 35$ s, resulting in changes in other WMR parameters. Three payload-change scenarios are considered, in which additional loads of 10%, 30%, and 50% of the original WMR mass are applied. Details of the parameter changes are presented in Table 5.

The adaptive observer parameters were tuned using particle swarm optimization (PSO) based on data from Scenario 3. The optimized parameters, summarized in Table 6, were then applied to all parameter-change scenarios. The WMR inputs were designed to ensure persistency of excitation, as illustrated in Figure 9. The parameter estimation results for all scenarios are presented in Figure 10.

In this simulation, the parameter estimation gains converge within approximately 15 s (300 data points) and re-converge at around 35 s (700 data points) after the parameter change. Detailed simulation results, including the RMSE and MAPE values, are summarized in Table 7. The performance metrics reported in the table are computed using the last 150 data points after the steady-state condition is reached. The MAPE values for all parameter estimates across all scenarios remain below 6.1%, indicating that WyNDA is capable of accurately estimating piecewise time-varying parameters.

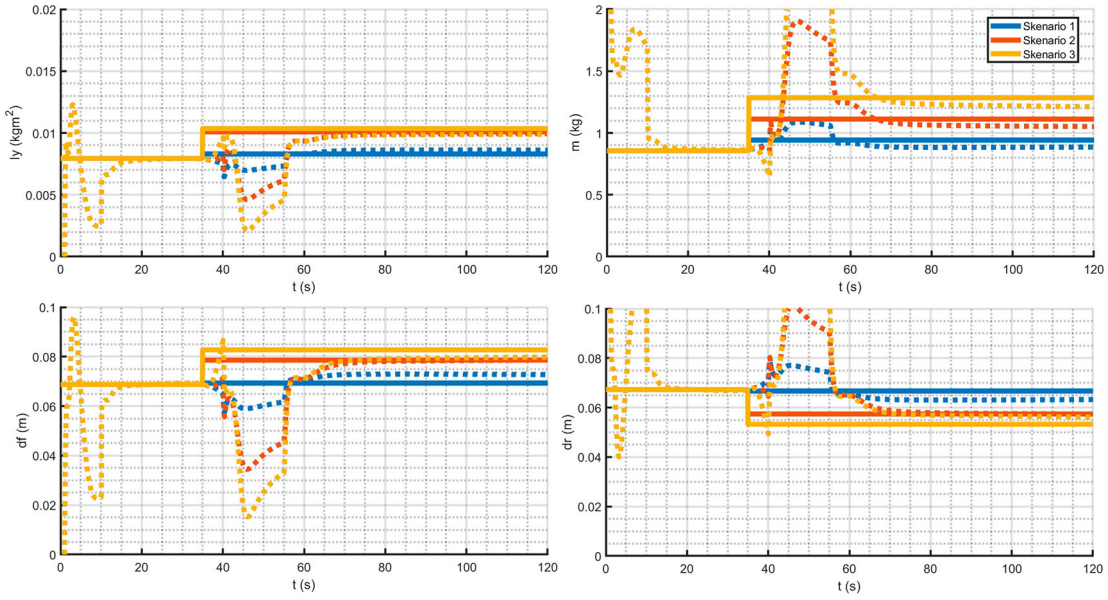


Figure 10. Parameter estimation results for the WMR parameter-change simulation.

Table 7. Estimated WMR parameters and corresponding RMSE and MAPE for the parameter-change simulation.

Parameter	Reference Value	Estimated Value	RMSE	MAPE
Scenario 1				
m (kg)	0.942	0.885	0.057	6.02%
I (kgm^2)	8.3×10^{-3}	8.6×10^{-3}	3.11×10^{-4}	3.74%
d_f (m)	6.93×10^{-2}	7.28×10^{-2}	3.5×10^{-3}	5.02%
d_r (m)	6.67×10^{-2}	6.32×10^{-2}	3.5×10^{-3}	5.21%
Scenario 2				
m (kg)	1,112	1.05	0.060	5.42%
I (kgm^2)	10.08×10^{-3}	9.9×10^{-3}	1.41×10^{-4}	1.4%
d_f (m)	7.86×10^{-2}	7.86×10^{-2}	2.18×10^{-5}	0.024%
d_r (m)	5.74×10^{-2}	5.74×10^{-2}	2.18×10^{-5}	0.033%
Scenario 3				
m (kg)	1.284	1.21	0.071	5.54%
I (kgm^2)	10.35×10^{-3}	9.9×10^{-3}	4.66×10^{-4}	4.5%
d_f (m)	8.275×10^{-2}	7.98×10^{-2}	2.9×10^{-3}	3.56%
d_r (m)	5.325×10^{-2}	5.62×10^{-2}	2.9×10^{-3}	5.54%

Table 8. Adaptive Observer Initial Conditions (Experiment).

Parameter	Value	Parameter	Value	Parameter	Value
$\boldsymbol{E_x}(0 \mid -1)$	$0.01\boldsymbol{I}$	$\boldsymbol{T_x}(1)$	$1 \times 10^5 \boldsymbol{I}$	$\boldsymbol{\Pi}(0 \mid 0)$	0
$\boldsymbol{E_\theta}(0 \mid -1)$	1.35×10^4	$\boldsymbol{T_\theta}(1)$	$1 \times 10^5 \boldsymbol{I}$	$\mu_{\boldsymbol{x}}$	0.7
				$\mu_{\boldsymbol{\theta}}$	0.995

4.2. Experimental results

The experimental testbed used in this study is shown in Figure 11. The WMR's speed and steering angle were measured using a tachometer and potentiometer, while its xy position and yaw angle were obtained via an indoor GPS. Lateral velocity and yaw rate were subsequently derived by differentiating the y-position and yaw angle, respectively. The adaptive observer parameters were tuned using PSO, with the resulting values summarized in Table 8.

The input signals used in this experiment are illustrated in Figure 12, while the resulting WMR trajectory is presented in Figure 13. In the latter, the black line corresponds to the WMR model, while the red dots represent the trajectory estimated via the WyNDA framework. A noticeable deviation occurs between the model

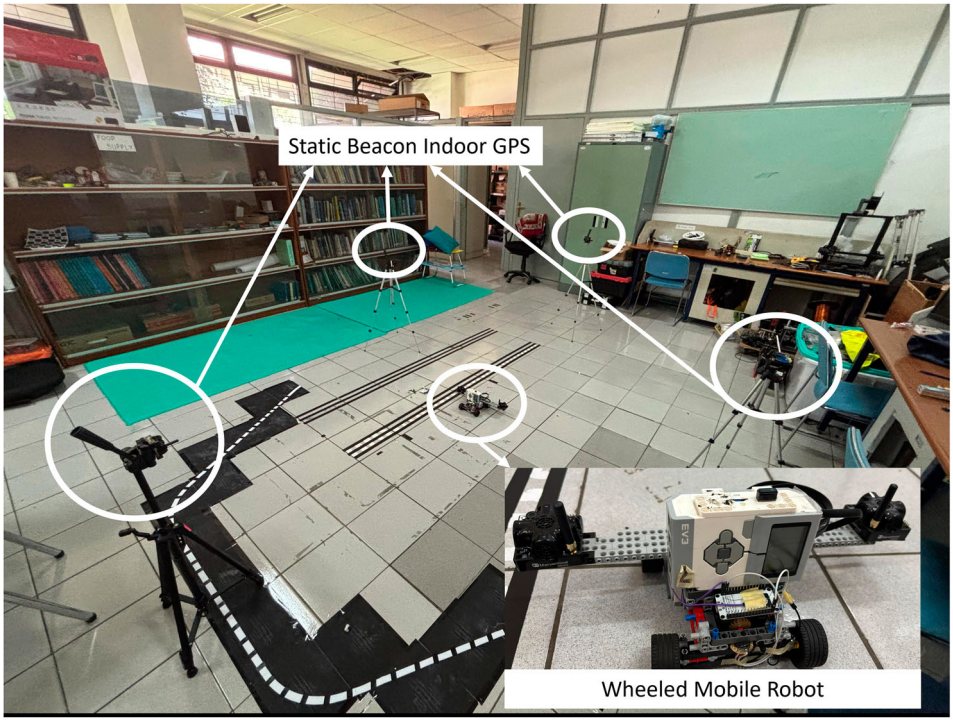


Figure 11. Experimental testbed used for vehicle parameter estimation.

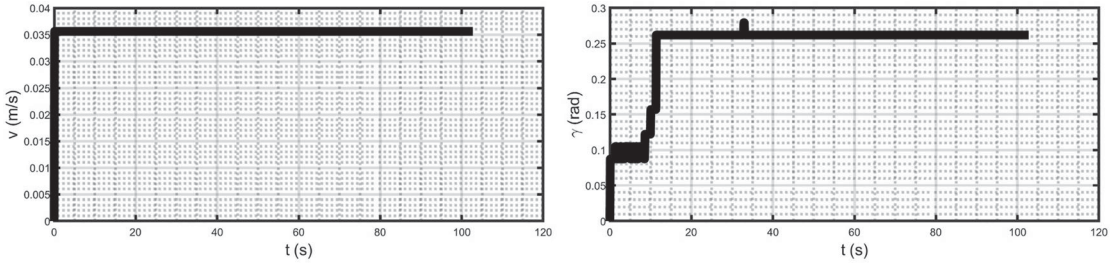


Figure 12. Input signals during experiment.

Table 9. Steady state estimated WMR parameters from the experiment.

Parameter	m (kg)	I (kgm^2)	d_f (m)	d_r (m)
3D Model	0.8561	7.9×10^{-3}	6.87×10^{-2}	6.73×10^{-2}
WyNDA	0.7642	10.09×10^{-2}	6.63×10^{-2}	6.97×10^{-2}
SINDy	0.525	34.57×10^{-2}	6.38×10^{-2}	7.22×10^{-2}

and the WyNDA estimation at the start of the experiment; however, the trajectories gradually converge and demonstrate strong agreement as time progresses.

Figure 14 illustrates the parameter estimation results obtained from experimental testing, while the steady-state estimates are summarized in Table 9. In Figure 14, the blue lines indicate the reference parameters used for comparison. These references are derived from a 3D model rather than direct physical measurements; thus, they should not be regarded as exact ground truth. Nevertheless, for parameters such as vehicle mass, the 3D model values are consistent with independent physical weighing, providing a reliable baseline for validation.

In contrast, parameters associated with rotational dynamics, such as the yaw moment of inertia, are more sensitive to modelling assumptions and mass distribution uncertainties. Consequently, these parameters may

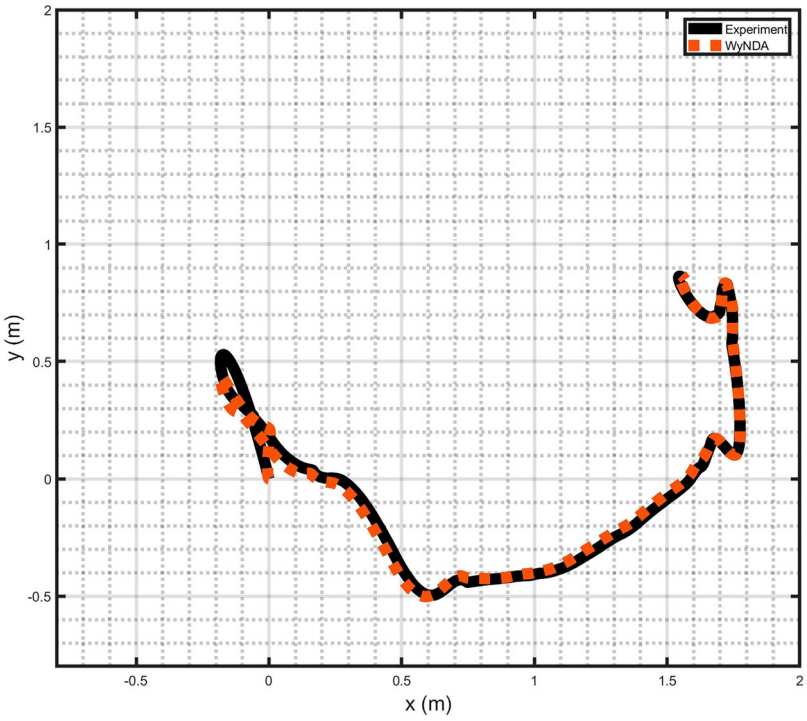


Figure 13. WMR trajectory during experiment.

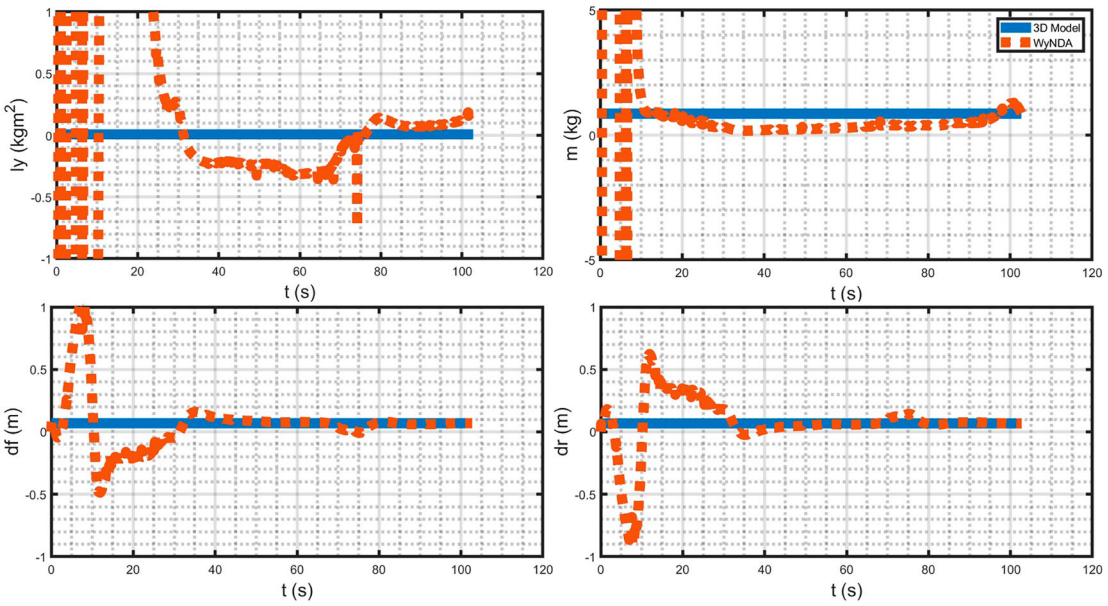


Figure 14. Estimated parameters of WMR throughout the experiment.

be less accurately represented by the 3D model. Since the true yaw moment of inertia is not directly measurable, the absolute accuracy of the WyNDA's estimation for this parameter cannot be conclusively validated using experimental data alone.

For completeness, the parameter estimation results obtained using SINDy are reported in Table 9 alongside those of WyNDA. The results indicate that, even when employing the full experimental dataset of 1000 points, the SINDy estimates exhibit larger deviations from the 3D-model reference values for mass and COG position than those produced by WyNDA. Specifically, WyNDA and SINDy yield a relative errors of 10.73% and 38.67%, respectively in estimating the vehicle mass.

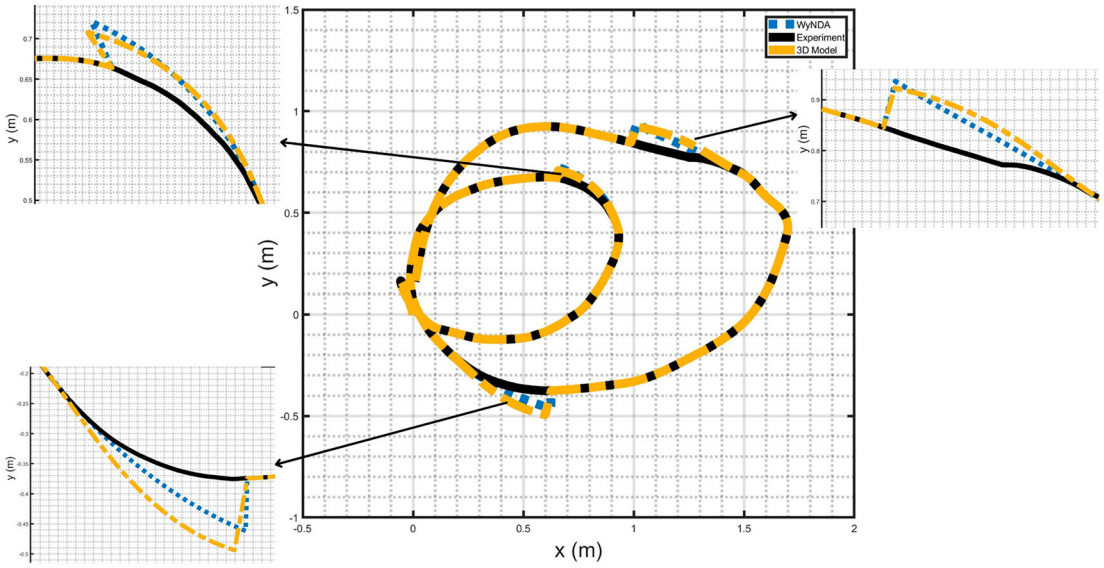


Figure 15. Simulation of digital twin compensating unavailable sensor measurement.

These experimental results under constant parameter conditions indicate that WyNDA provides a more reliable estimate of the WMR's mass than SINDy. Furthermore, simulation results under piecewise time-varying parameter values demonstrate that WyNDA consistently identifies all target parameters with high accuracy. Collectively, these results highlight the robustness of WyNDA and its suitability for integration into digital twin frameworks, particularly in scenarios where direct measurement of internal system parameters is impractical.

4.3. Digital twin implementation

Parameter estimation is conducted to build a reliable digital twin model. In this study, the estimated parameters using WyNDA are utilized in the simulation of the digital twin's position. A specific scenario is considered in which, during certain time windows, the physical WMR loses its ability to measure or transmit its state data. When this occurs, the missing data is substituted with predictions generated by the digital twin model. To evaluate the accuracy of this substitution, the differences between the digital twin and physical WMR data are quantified using RMSE and MAPE.In this scenario, the physical WMR experiences data loss every 25 seconds, with each loss lasting for 5 seconds. For comparison, the same scenario is also simulated using the parameter values obtained from 3D modelling, allowing evaluation of the estimation accuracy under identical conditions.

The simulation results of the digital twin system are shown in Figure 15. The black line corresponds to the nominal model simulation, the blue dots represent the simulation output using parameters estimated by the WyNDA framework, and the yellow dashed line indicates the result obtained using the reference parameter. The performance of the digital twin, based on WyNDA-estimated parameters, is compared to the simulation using the reference parameters, and the differences are quantified using RMSE and MAPE, as presented in Table 10. As shown in the table, the digital twin constructed using parameters estimated by WyNDA exhibits better performance in y position estimation compared to the one based on modelled parameters. Although the x position estimation accuracy of WyNDA is slightly lower, the difference remains limited and does not significantly affect the overall performance. These results indicate that the parameters estimated by WyNDA are effective and suitable for constructing a reliable digital twin.

5. Conclusion

This study demonstrates the effectiveness of WyNDA for real-time parameter estimation of a wheeled mobile robot (WMR) under diverse operating conditions. Simulation results show that WyNDA accurately estimates constant parameters in the presence of measurement noise, achieving MAPE values below 3.5% for of the

Table 10. RMSE and MAPE comparison for digital twin based on WyNDA-estimated vs. 3D Model-estimated.

	RMSE		MAPE	
Position	WyNDA	3D Model	WyNDA	3D Model
x	0.019 m	0.01 m	2.84%	1.33%
y	0.038 m	0.057 m	6.00%	9.49%

mass, moment of inertia, and COG positions. Compared to SINDy under identical conditions, WyNDA consistently reaches 80% estimation accuracy after approximately 304–306 data samples across multiple noise realizations, whereas SINDy exhibits less consistent performance. Furthermore, WyNDA successfully trakcs piecewise time-varying parameters under various payload-change scenarios, with MAPE values remaining below 6.1%. Finally, experimental validation under constant parameter conditions indicates that WyNDA provides a more reliable estimate of the WMR's mass than SINDy, achieving a relative errors of 10.73% compared to 38.67%.

To evaluate the robustness of the digital twin, a periodic data-loss scenario was simulated, where the digital twin output was used to reconstruct missing state information. Performance evaluation using RMSE and MAPE metrics demonstrates that the WyNDA-based digital twin achieves high accuracy, with MAPEs of 2.84% and 6.00% for position in the *x* and *y* coordinates, respectively. This performance is slightly better than the baseline model and shows close agreement with the experimental data. These results confirm the suitability of the proposed WyNDA-based approach for real-time digital twin applications under incomplete sensing conditions.

Future work will focus on extending the digital twin to represent more complex dynamic behaviours by including additional degrees of freedom beyond *xy* translation and yaw rotation. This includes modelling heaving, rolling, and pitching motions to better reflect real-world operating conditions. In parallel, parameter estimation will be enhanced to account for coupled dynamics involving tyre compliance, suspension effects, and many more. Theoretical development is also needed to establish the necessary and sufficient conditions for convergence and stability of the estimated parameters under noisy and uncertain conditions. Finally, real-world validation on physical WMR platforms, including tests with sensor degradation and communication delays, will be essential to assess the framework's practical applicability and performance.

Author contributions

CRediT: **Shinta R. Kosmaga:** Conceptualization, Data curation, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing; **Augie Widyotriatmo:** Conceptualization, Methodology, Supervision, Writing – review & editing; **Estiko Rijanto:** Conceptualization, Investigation, Methodology, Resources, Supervision, Writing – review & editing; **Agus Hasan:** Conceptualization, Funding acquisition, Methodology, Supervision, Validation, Writing – review & editing

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research is partially funded by Equinor ASA with project number 101593114; and RIIM Program of the Badan Riset dan Inovasi Nasional (National Research and Innovation Agency), Indonesia.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability statement

The codes and data are available here: https://doi.org/10.5281/zenodo.18316060.

References

Abe, M. (2015). Chapter 3 – Fundamentals of vehicle dynamics. In M. Abe (Ed.), *Vehicle handling dynamics* (2nd ed., pp. 45–107). Butterworth-Heinemann.

Brunton, S. L., Proctor, J. L., & Kutz, J. N. (2016). Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, *113*(15), 3932–3937.

Champion, K., Lusch, B., Kutz, J. N., & Brunton, S. L. (2019). Data-driven discovery of coordinates and governing equations. *Proceedings of the National Academy of Sciences*, *116*(45), 22445–22451.

Dong, G., Che, G., Tian, M., Zhao, H., & Gao, B. (2021). Vehicle state estimation based on recurrent neural network. In *2021 40th Chinese Control Conference (CCC)* (pp. 2669–2673). IEEE.

Fuller, A., Fan, Z., Day, C., & Barlow, C. (2020). Digital twin: Enabling technologies, challenges and open research. *IEEE Access*, *8*, 108952–108971.

Gräber, T., Lupberger, S., Unterreiner, M., & Schramm, D. (2019). A hybrid approach to side-slip angle estimation with recurrent neural networks and kinematic vehicle models. *IEEE Transactions on Intelligent Vehicles*, *4*(1), 39–47.

Hasan, A. (2024a). Discovering state-space representation of dynamical systems from noisy data. *IEEE Access*, *12*, 108744–108754.

Hasan, A. (2024b). WyNDA: A method to discover mathematical models of dynamical systems from data. *MethodsX*, *12*, 102625.

Hasan, A., & Hilmi, M. (2024). Discovering governing equations of robots from data. In *2024 IEEE International Conference on Real-Time Computing and Robotics (RCAR)* (pp. 206–211). IEEE.

Hong, S., Lee, C., Borrelli, F., & Hedrick, J. K. (2015). A novel approach for vehicle inertial parameter identification using a dual Kalman filter. *IEEE Transactions on Intelligent Transportation Systems*, *16*(1), 151–161.

Jin, X., Yang, J., Xu, L., Wei, C., Wang, Z., & Yin, G. (2023). Combined estimation of vehicle dynamic state and inertial parameter for electric vehicles based on dual central difference Kalman filter method. *Chinese Journal of Mechanical Engineering*, *36*(91), 1–16.

Jin, X., Yang, J., Zhu, T., Wang, J., & Yin, G. (2019). Real-time estimation of inertial parameter for lightweight electric vehicle using dual Kalman filter. In *2019 3rd Conference on Vehicle Control and Intelligence (CVCI)* (pp. 1–5). IEEE.

Kim, M., & Jung, D. (2024). Vehicle mass estimation via practical supervisory artificial neural networks using perturbed engine torque and acceleration inputs. *Applied Sciences*, *14*(23). MDPI.

Kosmaga, S. R., Rijanto, E., Widyotriatmo, A., & Hasan, A. (2024). Online vehicle parameter identification using wide-array of nonlinear dynamics approximation. In *2024 14th Asian Control Conference (ASCC)* (pp. 2485–2490). IEEE.

Kosmaga, S. R., Widyotriatmo, A., & Rijanto, E. (2025). Digital twin of autonomous vehicle using time-varying vehicle parameters estimation. In *2025 9th International Conference on Instrumentation, Control, and Automation (ICA)* (pp. 207–212). IEEE.

Li, W., Li, H., Xu, K., Huang, Z., Li, K., & Du, H. (2021). Estimation of vehicle dynamic parameters based on the two-stage estimation method. *Sensors*, *21*(11), 1–17.

Oei, M., & Sawodny, O. (2022). Vehicle parameter estimation with Kalman filter disturbance observer. *IFAC-PapersOnLine*, *55*(27), 497–502.

Piromalis, D., & Kantaros, A. (2022). Digital twins in the automotive industry: The road toward physical-digital convergence. *Applied System Innovation*, *5*(4), 1–12.

Qin, W. B., Zhang, Y., Takács, D., Stépán, G., & Orosz, G. (2022). Nonholonomic dynamics and control of road vehicles: Moving toward automation. *Nonlinear Dynamics*, *110*(3), 1959–2004.

Rajamani, R. (2012). Lateral vehicle dynamics. In *Vehicle dynamics and control* (pp. 15–46). Springer.

Veledar, O., Damjanovic-Behrendt, V., & Macher, G. (2019). Digital twins for dependability improvement of autonomous driving. In A. Walker, R. V. O'Connorand, & R. Messnarz (Eds.), *Systems, software and services process improvement* (pp. 415–426). Springer International Publishing.

Verma, R., Del Vecchio, D., & Fathy, H. K. (2008, October). Longitudinal vehicle dynamics scaling and implementation on a HIL setup. In *Asme 2008 Dynamic Systems and Control Conference, Parts A and B* (pp. 979–986). ASME.

Xu, Y., Deng, B., & Xu, G. (2015). Estimation of vehicle states and road friction based on DEKF. In *2015 6th International Conference on Power Electronics Systems and Applications (PESA)* (pp. 1–7). IEEE.

Yu, Z., Hou, X., Leng, B., & Huang, Y. (2022). Mass estimation method for intelligent vehicles based on fusion of machine learning and vehicle dynamic model. *Autonomous Intelligent Systems*, *2*(4), 1–10.

Zarringhalam, R., Rezaeian, A., Melek, W., Khajepour, A., Chen, S. K., & Moshchuk, N. (2012). A comparative study on identification of vehicle inertial parameters. In *2012 American Control Conference (ACC)* (pp. 3599–3604). IEEE.